

TBD*

Dan Bernhardt¹, Kostas Koufopoulos², and Giulio Trigilia³

¹*University of Illinois and University of Warwick*

²*University of York*

³*University of Rochester*

September 17, 2020

Abstract

TBD

Key words: TBD

JEL classification: TBD

*All errors are our own.

1 Introduction

2 The Model

We consider an economy that lasts three dates, $t = 0, 1, 2$. There are three types of agents: firms, banks and arm's-length investors. The firm seeks financing for a project that requires a fixed investment $I > 0$. The expected return of the project depends positively on the unobservable effort level $e \in [0, \infty)$ chosen by the firm, which has unit cost $c(e) = e$. The liquidation value of the firm's project at $t = 1$ is L . If it is not liquidated, the project's payoff at $t = 2$ depends on the state of nature $s = \{G, B\}$ that realizes at $t = 1$. Specifically, if $s = G$ the payoff is X with probability one. If $s = B$, instead, the payoff is X with probability $q \in (0, 1)$, and zero otherwise. Define $p(e) := \Pr.[s = G|e]$.¹ Following [Rajan \(1992\)](#), we make the following assumptions on parameters:

A1: Liquidation is optimal if and only if $s = B$: $X > I > L > qX$.

A2: The probability of success is increasing ($p'(e) > 0$) and concave ($p''(e) < 0$).

A3: The marginal benefit of effort satisfies: $p'(\infty) = 0$; $p'(0) = \frac{1}{\epsilon}$ for a small $\epsilon > 0$.

The firm has no retained earnings at $t = 0$, but it has access to a perfectly competitive banking sector. Banks can finance the project at $t = 0$ and, conditional on the contract being accepted, choose whether to monitor the firm or not. If a bank monitors, it learns the state of the realized state of the world at $t = 1$, at a fixed cost $c \geq 0$. In addition to bank capital, the firm can also raise funding from a large number of arm's-length investors. Unlike the bank, these investors are incapable of monitoring the firm, and therefore they do not learn the state of the world at $t = 1$.

2.1 Constrained Best Solution when the monitoring cost is zero

We first consider the planner's problem when the monitoring cost is zero. In this case, the planner solves the following optimization program:

$$\max_e p(e) (X - R_{c=0}) - e \tag{1}$$

where $R_{c=0}$ is the repayment specified by the planner for the lender. The FOC is necessary and sufficient for an optimum—given our assumptions—and can be written as:

¹For comparative statics purposes, in [Rajan \(1992\)](#) p is a function of both effort and an exogenous quality parameter $\alpha \in [0, 1]$ which is common knowledge ex ante (i.e., $p \equiv p(e, \alpha)$). Including this in our model would complicate the notation without making any qualitative difference to the results.

$$p'(e_{c=0}^*) = \frac{1}{(X - R_{c=0})} \quad (2)$$

Notice that the efficient solution requires continuation of the project if and only if the state at $t = 1$ is revealed to be good. Otherwise, the project should be liquidated. However, the firm's payoff is zero upon liquidation, while it is positive upon continuation. Thus, if the firm controls the liquidation decision, it would always inefficiently continue in the bad state. In addition, observe that arm's length investors could not influence the decision, given that by assumption they are not able to observe the state of nature at $t = 1$. As a result, the arm's length contract is clearly suboptimal, as stated in Lemma 1:

Lemma 1. *Any arm's length contract cannot credibly deter continuation in the bad state at $t = 1$. Therefore, it cannot implement the planner's solution.*

According to the analysis in [Rajan \(1992\)](#), the constrained best investment level could not be implemented via bank financing either, despite perfect competition. The argument can be summarized as follows. Suppose that the bank specifies a long-term straight debt contract, where $D_{LT,b}$ is the face value of debt. In order to break even, banks must be guaranteed that at $t = 1$ there is continuation if and only if the state of nature turns out to be good. Otherwise, there must be liquidation. However, the firm would always continue, for the reason explained above. So, when the bad state realizes at $t = 1$, the bank should bribe the firm in order to provide her enough incentives to liquidate. They would bargain over the surplus $L - qX > 0$, and the final allocation will be a function of both the bargaining power across agents, and the structure of the renegotiation game. Because this redistribution is anticipated ex-ante, and since it has the adverse effect of increasing the firm's payoff in the bad state, it reduces the firm's incentives to exert effort. It follows immediately that any (straight-debt) long-term banking contract would not implement the planner's solution either. This is summarized in our Lemma 2.

Lemma 2. *Under any long-term, straight-debt contract issued by banks, the firm's payoff in the bad state is strictly greater than zero, generically—i.e., if its bargaining power is not exactly zero. Therefore, such contract cannot implement the planner's solution.*

Now, let's consider a short-term, straight debt banking contract. In this case, the bank requires repayment of $D_{ST,b}$ at $t = 1$. Clearly, in equilibrium we will have $D_{ST,b} = L$. Abstracting from competition between banks and investors at $t = 1$ (which complicates the setup without changing the essence of the argument), it is evident that the contract achieves the goal of forcing liquidation in the bad state. However, a problem arises in the

good state, where the firm enjoys the expected surplus $X - L > 0$. Depending again on the bargaining power of the agents and on the specification of the bargaining game, this surplus will be divided between them arbitrarily. Once again, this has the adverse effect of reducing the firm's incentives to exert effort ex-ante, which implies that the short-term contract does not implement the planner's solution either. This sketches the reasoning that justifies the 'natural' cost of banking argument outlined in [Rajan \(1992\)](#).

Lemma 3. *Under any short-term, straight-debt contract issued by banks, the firm's payoff in the good state is strictly lower than that required to implement the planner's solution.*

At this point, one may conclude that the planner's solution is not implementable in this setting. However, we note that these inefficiencies critically depend on the restriction to plain vanilla debt contracts, which is both not plausible empirically, and not micro-founded theoretically in this environment. We now provide a simple, renegotiation-proof debt contract that banks can issue and which implements the planner's solution for all parameter values. Specifically, suppose that the bank issues debt with a covenant that shifts the control rights over the liquidation decision from the firm to the bank, unconditionally on the state of nature.² We now show that this implements the planner's solution.

The maximization problem for the firm under this debt contract is given by:

$$\max_e p(e)(X - D_{b|c=0}) - e \quad (3)$$

where $D_{b|c=0}$ denotes the repayment required by the bank offering the debt with covenant contract, conditional on monitoring costs being zero. Notice that in writing this problem, we have assumed that the bank will always continue the project in the good state, and it will always liquidate it in the bad state. Moreover, we assumed that the contract is not renegotiated. These two claims are stated and proved in [Lemma 4](#).

Lemma 4. *Under the debt-with-covenant contract, the bank always makes the efficient liquidation decision at $t = 1$: it always liquidates if $s = B$, and it continues if $s = G$. Moreover, this contract is renegotiation-proof.*

Proof. First, observe that in the bad state at $t = 1$ liquidation is efficient, and the bank gets the entire liquidation value L . Continuation is a strictly dominated action because $L > qX$. Second, note that it is not possible to renegotiate in the bad state. This is

²Notice that no covenant could be made conditional on the state in our setting, because the state is observable but non-verifiable by a court.

because the firm has no control rights over the liquidation decision, and so it cannot influence the bank in any possible way. As for the bank, once the bad state is reached at $t = 1$, the maximum amount it can obtain is precisely L , and therefore any threat it could make would not be credible.

Now consider the good state. Here, the bank could try to extract part of the surplus $X - L > 0$ from the firm by renegotiating the terms of the contract, under the threat of liquidation. However, is this threat credible once we reached the good state? The answer is negative, because we know that $D_{b|c=0} > L$, otherwise the bank's participation constraint would not be satisfied. Thus, the firm can safely refuse to renegotiate because the bank will never terminate the project. $\square$

Now we can proceed to the solution of the previous maximization problem. The participation constraint for the bank in this case is given by:

$$p(e)D_{b|c=0} + (1 - p(e))L \geq I \quad (4)$$

Because of competition, this constraint binds and we get the following equation:

$$p'(e_{b|c=0}^*) = \frac{1}{X - D_{b|c=0}} = \frac{1}{X - L - \frac{I - L}{p(e_{b|c=0}^*)}} \quad (5)$$

As $L < I$, we know that $p'(e_{b|c=0}^*) > p'(e_{FB}^*)$, which implies that $e_{b|c=0}^* < e_{FB}^*$ by assumption **A2**. Thus, the effort exerted does not reach the First-Best because of limited liability. However, the solution implements the Second-Best, as it features (1) the efficient liquidation decision at $t = 1$; and (2) zero profits for the bank ex ante. It follows immediately that accepting this contract is a dominant strategy for the firm, and therefore no arm's length contract can be preferred to the banking contract, as stated in Proposition 1:

Proposition 1. *If $c = 0$, debt-with-covenant implements the planner's solution (i.e., the Second-Best). In particular, it strictly dominates any 'pure-debt' contract offered by banks, as well as any arm's length contract.*

Having established this first result, we can restrict attention to this contract when considering bank financing, and we can study the case of positive monitoring costs.

3 Solutions with monitoring costs

The economic explanations of why banks have an advantage in monitoring firms range from economies of scope or scale, to other dimensions discussed in the introduction. However, all theories share the idea that monitoring does not come free of costs. Otherwise, the existence of uninformed investors (what we call arm's length financiers) would become problematic. Therefore, in what follows, we relax the assumption of no monitoring costs and we analyse the case where $c > 0$.

As we are interested in the case where risk-less debt cannot be issued, we only characterize the Second-Best planner's problem, assuming that $L < I$.

3.1 Second-Best solution

The planner's problem now is more complex, in that it express a trade off between informed and uninformed financing. The trade off for the firm can be described as follows:

1. The banking contract has the positive feature of a higher expected rate of success, because of the higher probability of the good state realizing at $t = 1$. This translates into a lower expected repayment. However, it has the negative feature of incurring the monitoring cost;
2. On the contrary, the arm's length contract has the pros. of avoiding the monitoring cost. However, it has a lower probability of success ex-ante and hence the repayment in case of success is higher.

Intuitively, the quality parameter α will become a crucial determinant in the choice between banks and market investors.

The planner's problem can be stated as follows:

$$\max (\Pi_{NM}^*(\alpha), \Pi_M^*(\alpha))$$

where

$$\Pi_{NM}^*(\alpha) \equiv \max_e [p(e_{NM}, \alpha)(X - R_{NM}) + (1 - p(e_{NM}, \alpha))q(X - R_{NM}) - e_{NM}]$$

and

$$\Pi_M^*(\alpha) \equiv \max_e [p(e_M, \alpha)(X - R_M) - e_M]$$

Here the superscripts ‘NM’ and ‘N’ stand for ‘No Monitoring’ and ‘Monitoring’, respectively.

For any given project’s quality α , the planner first compares the maximum expected payoff from the firm perspective under no monitoring and monitoring, then she chooses the highest one.

Under no monitoring, we know from Lemma 1 that the firm always continues at $t = 1$. Therefore the feasibility constraint faced by the planner is:

$$R_{NM} = \frac{I}{p(e_{NM}, \alpha) + (1 - p(e_{NM}, \alpha))q}$$

Substituting it inside the objective function gives us the following FOC:

$$p_1(e_{NM}^*, \alpha) = \frac{1}{\left(X - I - \frac{(1 - p_{NM}^*)(1 - q)I}{p_{NM}^* + (1 - p_{NM}^*)q}\right)(1 - q)}$$

On the contrary, if there is monitoring the optimal liquidation decision is conditional on the realized state of nature: if the state is good there must be continuation, whereas if the state is bad there is liquidation. Here, the feasibility constraints will be:

$$pR_M = \frac{I - (1 - p(e_M, \alpha))L + c}{p(e_M, \alpha)}$$

In the same fashion as before, we find the following FOC:

$$p_1(e_M^*, \alpha) = \frac{1}{X - L - \frac{(I - L) + c}{p_M^*}}$$

The unique solution to each problem is given by the fixed point of equations (6) and (8), respectively.

We can rewrite the expected payoffs as follows:

$$\Pi_{NM}^*(\alpha) = p_{NM}^*X + (1 - p_{NM}^*)qX - I$$

and

$$\Pi_M^*(\alpha) = p_M^* X + (1 - p_M^*) L - (I + c)$$

The last stage of the planner's problem is going to be the comparison between $\Pi_M^*(\alpha)$ and $\Pi_{NM}^*(\alpha)$ for every project's quality α . Whenever $\Pi_M^*(\alpha) \geq \Pi_{NM}^*(\alpha)$, the planner is going to choose the option of monitoring - and vice versa.

3.2 Implementation of the optimal contract with positive monitoring costs

3.2.1 Arm's length optimal contract

As we assumed that learning of the state of nature at $t = 1$ is conditional on monitoring, then it is clear that whenever the bank does not monitor, it can offer at best the optimal contract feasible for arm's length financing. Therefore, we start by characterizing the arm's length optimal contract.

Observe that the uninformed investor has exactly the same information set at $t = 0$ and at $t = 1$. Hence, without loss of generality, we can restrict attention to Long-Term contracts. Suppose the contract specifies that the firm borrows I at $t = 0$, and has to repay R_a at $t = 2$.

The firm solves the following problem:

$$\max_e p(e, \alpha)(X - R_a) + (1 - p(e, \alpha))q(X - R_a) - e$$

Notice that it is exactly the planner's problem for the case of no monitoring (NM).

The FOC for this problem is given by:

$$p_1(e_a^*, \alpha) = \frac{1}{(X - R_a)(1 - q)}$$

In order to back up R_a , we notice that it must be the case that the expected return for the investor from financing the project is greater or equal to the cost of financing it. Moreover, by perfect competition we know that the participation constraint will be binding in equilibrium. Hence we get:

$$R_a = \frac{I}{p(e_a^*, \alpha) + (1 - p(e_a^*, \alpha))q}$$

A quick comparison shows that this equation has the same structure of equation (5) - i.e. the planner's feasibility constraint for the no monitoring case.

Substituting equation (3) in the FOC we get:

$$p_1(e_a^*, \alpha) = \frac{1}{\left(X - I - \frac{(1 - p_a^*)(1 - q)I}{p_a^* + (1 - p_a^*)q}\right)(1 - q)}$$

where we define $p_a^* \equiv p(e_a^*, \alpha)$. For relatively high quality projects a solution exists, and if it does it is unique³. The solution highlight how arm's length debt can implement the Second-Best in the no monitoring case.

3.2.2 Bank's optimal contract

Recall that if the bank does not monitor, it can offer at best the contract specifying repayment R_a , and we consider it as equivalent to an arm's length investor. Hence, from now onwards (without loss of generality) we suppose that banks always monitor.

Unlike the analysis in previous sections, now it becomes more expensive for banks to offer credit, hence in order to break even they require a higher repayment in case of success. The zero profit condition for the bank is now:

$$p(e, \alpha)D_{b|c>0} + (1 - p(e, \alpha))L - c \geq I$$

Again, the condition is equivalent to the planner's feasibility constraint for the monitoring case.

By plugging it inside the same FOC as before, we get:

$$p_1(e_{b|c>0}^*, \alpha) = \frac{1}{X - D_{b|c>0}} = \frac{1}{X - L - \frac{(I - L) + c}{p_{b|c>0}^*}}$$

where the only new term in the denominator is $c/p_{b|c>0}^* > 0$. A simple comparison between equations (4) and (7) shows that $p_1(e_{b|c>0}^*, \alpha) > p_1(e_{b|c=0}^*, \alpha)$, which implies: $e_{b|c>0}^* < e_{b|c=0}^*$. Intuitively, the more costly it is for the bank to provide credit, the lower the wealth at stake for the firm in starting a given project, the lower the effort she will supply.

A quick comparison of this solution with the planner's problem solved in (5.1) shows

³Uniqueness is guaranteed by (A2) and (A3). Notice that: $(X - R_a)(1 - q_B) < (X - L)$ for all $q_j \in [0, 1]$, because $R_a > L$ and $(1 - q_B) \in (0, 1)$. Therefore, by (A2) we have that $e_a^* < e_{FB}^*$.

that the banking contract implements the Second-Best for the monitoring case.

4 Comparative statics

Now that we showed how the Second-Best can be implemented by the availability of both banks and arm's length financing, we characterize the firmial choice between the two. In particular, we are interested in showing how firm's quality (α) and its amount of tangible assets L are going to affect the choice. The results are summarized in the following propositions:

Proposition 2. *There exists a threshold for project's quality $\hat{\alpha}$ such that:*

1. *If $\alpha < \hat{\alpha}$, then the planner chooses not to monitor, and the required repayment is R_{NM}^* ;*
2. *If $\alpha \geq \hat{\alpha}$, then the planner chooses to monitor, and the required repayment is R_M^* .*

Proof. Consider the two effects of a change in α on the probability of success $p_{b|c>0}^*$. The first is the direct effect, which is positive by assumption (A2). The second one is indirect, and depends on how the optimal effort level is affected by a change in α , *caeteris paribus*. Since a higher monitoring cost implies a higher success probability, by equation (13) we know that it reduces the repayment required by the bank to break even. But this implies that the effort exerted increases, which has a positive effect on the probability of success.

As a result, we know that when project's quality increases, the probability of success increases as well.

Taking the derivative of bank's participation constraint with respect to the monitoring cost we get:

$$\frac{\partial D_{b|c>0}}{\partial c} = \frac{1}{p_{b|c>0}^*}$$

Hence, a higher probability of success reduces the change in the required repayment for a given shift in the monitoring cost.

Since the change in the repayment is smaller for higher quality firms, also the fall in their optimal effort exerted will be smaller, compared to low quality firms. Hence, the higher the quality, the more likely it is for the firm to prefer the banking contract. $\square$

5 Characterization of firmial choice

Now that we derived the optimal contracts, we proceed to the characterization of firm's choice between them.

The expected payoff for the firm with bank financing is:

$$\Pi_b^* = p_{b|c>0}^* X + L (1 - p_{b|c>0}^*) - (I + c)$$

while the correspondent payoff with arm's length financing is:

$$\Pi_a^* = p_a^* X + (1 - p_a^*) qX - I$$

An firm will chose bank financing if and only if $\Pi_b^* \geq \Pi_a^*$, which is equivalent to:

$$(p_{b|c>0}^* - p_a^*) X - p_{b|c>0}^* L + p_a^* qX \geq c - qX - L$$

Given our previous discussion, we know that $p_{b|c>0}^*$ is the probability of the project succeeding given that the Second-Best effort level is exerted by the firm. Moreover, we argued that $p_{b|c>0}^* > p_a^*$ as no arm's length contract can implement the Second-Best (due to inefficient continuation in the bad state at $t = 1$). Hence, for every possible project's quality α , we have: $(p_{b|c>0}^* - p_a^*) X > 0$. We also know by (A1) that $L > qX$, but this implies that $(p_a^* qX - p_{b|c>0}^* L) < 0$. By (A2), we also know that the probability of success is increasing in project's quality.

Proposition 3. *There exists a cutoff value for project's quality $\hat{\alpha}$, such that for all projects with $\alpha > \hat{\alpha}$ arm's length financing is preferred to bank financing, while for all $\alpha \leq \hat{\alpha}$ bank financing is preferred to arm's length financing.*

Proof. Taking the derivative of the expected payoff functions for the firm we get that:

$$\begin{aligned} \bullet \quad \frac{\partial \Pi_b}{\partial \alpha} &= (X - L) \left[\frac{\partial p_{b|c>0}^*}{\partial e_{b|c>0}^*} \frac{\partial e_{b|c>0}^*}{\partial \alpha} + \frac{\partial p_{b|c>0}^*}{\partial \alpha} \right]; \\ \bullet \quad \frac{\partial \Pi_a}{\partial \alpha} &= X (1 - q) \left[\frac{\partial p_a^*}{\partial e_a^*} \frac{\partial e_a^*}{\partial \alpha} + \frac{\partial p_a^*}{\partial \alpha} \right]. \end{aligned}$$

Moreover, implicitly differentiating the FOCs for the respective maximization problems (i.e. equations (6) and (7)) with respect to α , we find:

□

This result is in line with what Rajan (1992) and Diamond (1991) found, even though its source is different. In Rajan, the choice is determined by inefficient renegotiation ex-post, while here it is driven by monitoring costs. It is important to stress that the difference is not innocuous: in our case banks and arm's length financier

References

Rajan, R, "Insiders and Outsiders: The Choice between Informed and Arm's Length Debt," *Journal of Finance*, 1992, *47*, 1367–1400.